Piyush Kumar

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PROFESSIONAL SUMMARY

Backend-focused Software Developer with hands-on experience building RESTful APIs, database-driven applications, and AI-powered systems using Node.js, Express.js, TypeScript, and PostgreSQL. Experienced in API development, database integration, authentication, and building RAG applications using LLM APIs, embeddings, and vector databases.

EDUCATION

Indian Institute of Technology, Patna — *Bihar*
Bachelor of Science (BS) in Computer Science and Data Analytics | CGPA: 9.0 — 2025 – 2029

PROJECTS

Mindora | *AI Knowledge Workspace* — Website | Frontend | Backend

- Developed a NotebookLM-style application supporting ingestion of PDFs, web URLs, YouTube Videos, and plain text, enabling users to query their documents through an AI-powered chat interface.
- Implemented an end-to-end RAG pipeline using Mistral embeddings and Pinecone vector search, integrating Groq LLMs to generate source-grounded answers with inline citations.
- Built RESTful APIs using Node.js, Express.js, and TypeScript for document processing, notebook management, and persistent chat history; integrated Clerk authentication and PostgreSQL with Prisma for data management.
- Manually validated YouTube ingestion and question answering across 10–20 video URLs covering different topics, testing multiple questions per video for response relevance.
- **Technologies:** React, TypeScript, Node.js, Express.js, PostgreSQL, Prisma, Pinecone, Mistral Embeddings, Groq API, Supadata API

Multi-Agent Research System | *Automated Report Generation with Self-Critique* — GitHub

- Architected 4-stage multi-agent LangChain system, with tool-using search and reader agents feeding writer and critic chains, autonomously turning a topic into a structured, sourced research report with independent review.
- Built custom LLM-callable tools (Tavily web search, BeautifulSoup scraper) with safeguards: request timeouts, graceful error handling, boilerplate stripping (scripts, navigation, footers), and capped snippet/page lengths to fit prompt limits.
- Implemented grounded generation: the writer is limited to gathered research (no outside knowledge, invented facts, or fabricated citations), while a separate LLM critic scores each report out of 10 with strengths, weaknesses, verdict.
- Composed writer and critic as LCEL chains (prompt → LLM → parser), orchestrating all four stages sequentially over shared state, each consuming the prior output, with every model at temperature 0 for repeatable results.
- **Technologies:** Python, LangChain, Groq API, Tavily API, Streamlit

EventEra | *FullStack Event Booking Platform* — GitHub

- Developed a full-stack event booking platform using React.js, Node.js, Express.js, and MongoDB, enabling event discovery, ticket booking, and reservation management.
- Implemented JWT-based authentication, bcrypt password hashing, email OTP verification, and role-based access control (RBAC) to secure user accounts and restrict administrative operations.
- Designed RESTful APIs for event management, booking creation, cancellation, and approval, incorporating duplicate-booking validation and event seat availability tracking.
- Built an admin dashboard for event CRUD operations, booking approval, payment-status management, and booking analytics; integrated Nodemailer for automated OTP and booking confirmation emails.
- **Technologies:** React.js, Node.js, Express.js, MongoDB, Mongoose, Tailwind CSS, JWT, bcryptjs, Nodemailer, Axios

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, Python, SQL
Frontend Development: React.js, Tailwind CSS
Backend Development: Node.js, Express.js, RESTful APIs, JWT Authentication
Databases & ORM: MongoDB, PostgreSQL, Redis, Prisma ORM, Mongoose
AI & Agentic Systems: Generative AI, LangChain, LangGraph, Agentic AI Workflows
Tools & DevOps: Git, GitHub, GitHub Actions (CI/CD), Docker, Postman, Nginx, Vercel, Render